📘 AI-BFSI Internship Project Documentation

Project Executable Files & Execution – Subscription Waste Detector

1. Introduction

The successful execution of a software application depends on a properly configured environment, required dependencies, and a clear execution procedure.

The **Subscription Waste Detector** is a **Streamlit-based web application** built with Python. It processes uploaded expense records, detects recurring subscription payments, generates interactive visualizations, and provides AI-powered subscription optimization recommendations via the Groq API.

This document explains prerequisites, folder structure, installation, and execution steps for running the application locally.

2. Project Overview

Attribute	Description
Project Name	Subscription Waste Detector
Application Type	AI-Assisted Financial Analysis Web Application
Platform	Streamlit
Programming Language	Python
Development Environment	Visual Studio Code
Execution Mode	Local Web Application

3. Project Directory Structure

```
Subscription-Waste-Detector/

|── app.py
|── requirements.txt
|── README.md
|
├── data/
├── utils/
├── reports/
└── assets/
```

Directory Description

Directory/File	Purpose
`app.py`	Main application entry point
`requirements.txt`	Python package dependencies
`README.md`	Setup instructions and project info

`utils/`	Helper functions and processing modules
`data/`	Sample datasets (optional)
`reports/`	Generated reports or supporting files
`assets/`	Static resources (icons, images, etc.)

4. Software Requirements

Software	Purpose
Python	Application runtime
Visual Studio Code	Development environment
Streamlit	Web application framework
Google Chrome	Browser to access the application

5. Python Libraries

Library	Purpose
Streamlit	User interface
Pandas	Data processing
Plotly	Interactive charts
OpenPyXL	Excel file support
Groq SDK	AI recommendation integration

Dependencies are installed via `requirements.txt`.

6. Installation Procedure

1. **Clone or download** the project files.
2. **Navigate** to the project directory:

   bash

   ```
   cd Subscription-Waste-Detector
   ```

3. **Install dependencies**:

   bash

   ```
   pip install -r requirements.txt
   ```

4. **Configure Groq API key** in the project settings.
5. **Verify installation** by checking that all libraries are available.

7. Project Execution

Run the application with Streamlit:

bash

```
streamlit run app.py
```

Streamlit will start a local server and provide a URL, typically:

text
```
http://localhost:8501
```

Open this in a browser to access the application.

8. Application Workflow

```
Start Application
        |
        ▼
Launch Streamlit
        |
        ▼
Open Web Interface
        |
        ▼
Upload CSV / Excel
or
Manual Expense Entry
        |
        ▼
Process Expense Data (Pandas)
        |
        ▼
Detect Subscriptions
        |
        ▼
Generate Dashboard
        |
        ├────────▶ Plotly Charts
        |
        └────────▶ Groq API
                     |
                     ▼
          AI Recommendations
                     |
                     ▼
           Display Results
```

9. Input Requirements

- **File Uploads:** CSV or Excel (.xlsx) files
- **Manual Entry:** Expense description, category, amount, renewal date

10. Expected Output

- Processed expense records
- Detected recurring subscriptions
- Subscription summary
- Monthly spending analysis
- Interactive Plotly charts
- Renewal reminders

- AI-powered recommendations

11. Execution Validation

Execution is successful when:

- Streamlit launches without errors
- Web interface loads correctly
- CSV/Excel uploads are processed
- Manual entries are accepted
- Subscription detection completes
- Dashboard displays summaries and charts
- AI recommendations are generated via Groq API

12. Known Limitations

- Internet required for AI recommendations
- No authentication implemented
- No database storage (session-based only)
- Intended for local execution only

13. Future Improvements

Potential enhancements:

- Database integration
- User authentication
- Cloud deployment
- Multi-user support
- Email notifications
- Bank API integration

14. Conclusion

The Subscription Waste Detector is a lightweight Streamlit web application that runs locally with minimal setup. By installing dependencies, configuring the Groq API key, and launching via Streamlit, users can upload expense records, detect subscriptions, visualize spending, and receive AI-powered recommendations.